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Configuring NXLog for HostIP Field

For NXLog, some configuration is needed to create the `HostIP` field from the `Hostname` field. This configuration in NXLog enables the collection of the host or source IP address when Windows events are ingested from NXLog.

The NXLog version must be 5.5 or later.

To configure NXLog:

1. Import Extension Modules
2. Configure NXLog
3. Refer to the Sample NXLog Configuration File
4. Refer to the Alert Types Received

Import Extension Modules

To import two extension modules, Resolver (xm_resolver) and Syslog (xm_syslog), you can refer to the following NXLog documentation:

- https://docs.nxlog.co/refman/current/xm/resolver.html
- https://docs.nxlog.co/refman/current/xm/syslog.html

Then import the extension modules:

```
<Extension _resolver>
    Module     xm_resolver
</Extension>


<Extension syslog>
    Module    xm_syslog
</Extension>
```

Configuring NXLog

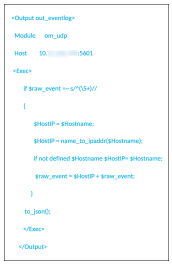
For rewriting and modifying logs, you can refer to the following NXLog documentation: https://docs.nxlog.co/userguide/configure/rewriting.html.

To configure NXLog to create the `HostIP` field:

1. In the input module, configure the listening host:
    - Use `parse_syslog()`. Parse the `$raw_event` field as either BSD Syslog (RFC 3164) or IETF Syslog (RFC 5424) format.
    - If there is more than one instance of xm_syslog, reference the module instance with `INSTANCENAME->parse_syslog()`, otherwise use `parse_syslog()`.

```
<Input tcp>
    Module     im_tcp
    ListenAddr  10.[redacted]  #hostip
    <Exec>
        syslog->parse_syslog();  # one instance can use "parse_syslog();"
    </Exec>
</Input>
```

2. In the output module, create the `HostIP` from the `Hostname` and rewrite `HostIP` to get the output:
    - Use `ipaddr name_to_ipaddr(string name)` to change the value of `Hostname` to the value of `HostIP`.



    The `HostIP` is added to the `raw_event` field to display the output.

Sample NXLog Configuration File

```
Panic Soft

define INSTALLDIR C:\Program Files\nxlog

#ModuleDir %INSTALLDIR%\modules
#CacheDir  %INSTALLDIR%\data
#SpoolDir  %INSTALLDIR%\data

define CERTDIR %INSTALLDIR%\cert
define CONFDIR %INSTALLDIR%\conf\nxlog.d

# Note that these two lines define constants only; the log file location
# is ultimately set by the `LogFile` directive (see below). The
# `MYLOGFILE` define is also used to rotate the log file automatically
# (see the `_fileop` block).
define LOGDIR %INSTALLDIR%\data
define MYLOGFILE %LOGDIR%\nxlog.log
```

```
# If you are not using NXLog Manager, disable the `include` line
# and enable LogLevel and LogFile.
include %CONFDIR%\*.conf

#LogLevel    INFO
#LogFile     %MYLOGFILE%

<Extension _syslog>
    Module  xm_syslog
</Extension>

# This block rotates `%MYLOGFILE%` on a schedule. Note that if `LogFile`
# is changed in managed.conf via NXLog Manager, rotation of the new
# file should also be configured there.
<Extension _fileop>
    Module  xm_fileop

    # Check the size of our log file hourly, rotate if larger than 5MB
    <Schedule>
        Every   1 hour
        <Exec>
            if ( file_exists('%MYLOGFILE%') and
                 (file_size('%MYLOGFILE%') >= 5M) )
            {
                 file_cycle('%MYLOGFILE%', 8);
            }
        </Exec>
    </Schedule>

    # Rotate our log file every week on Sunday at midnight
    <Schedule>
        When    @weekly
        Exec    if file_exists('%MYLOGFILE%') file_cycle('%MYLOGFILE%', 8);
    </Schedule>
</Extension>


<Extension syslog>
  Module    xm_syslog
</Extension>

<Extension _json>
  Module    xm_json
</Extension>
```

```
<Extension xml>
    Module      xm_xml
</Extension>
<Extension _resolver>
    Module      xm_resolver
</Extension>

<Input in_eventlog>
# For windows 2008/vista/7/8/2012/2012R2/2016 and latter use the following:
  Module      im_msvistalog
  ReadFromLast TRUE
  SavePos     TRUE
  Query     <QueryList> \
                                                    <Query Id="0"> \
                                                           <Select Path="Security">*
[System[(EventID=1100)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4768)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4769)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4771)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4616)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4104)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4624)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4625)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4634)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4647)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4648)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4656)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4688)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4719)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4720)]]</Select> \
                                                           <Select Path="Security">*
[System[(EventID=4722)]]</Select> \
```

```
                                                    <Select Path="Security">*
[System[(EventID=4723)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4724)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4725)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4726)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4727)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4728)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4729)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4730)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4731)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4732)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4733)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4734)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4735)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4737)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4738)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4739)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4740)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4741)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4742)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4743)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4744)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4745)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4748)]]</Select> \
```

```
                                                    <Select Path="Security">*
[System[(EventID=4749)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4750)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4753)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4754)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4755)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4756)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4758)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4759)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4760)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4763)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4764)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4767)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4776)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4778)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4783)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4800)]]</Select> \
                                                    <Select Path="Security">*
[System[(EventID=4801)]]</Select> \
                                                    <Select Path="System">*
[System[(EventID=7036)]]</Select> \
                                                    <Select Path="Setup">*
[System[(EventID=1)]]</Select> \
                                                    <Select
Path="Application">*[System[(EventID=18454)]]</Select> \
                                                    <Select
Path="Application">*[System[(EventID=18456)]]</Select> \
                                                </Query> \
                            </QueryList>
</Input>
```

```
<Input tcp>
    Module      im_tcp
    ListenAddr  10.xx.xx.xx


    <Exec>

        syslog->parse_syslog();


    </Exec>
</Input>

<Output out_eventlog>
  Module      om_udp
  Host        10.xx.xx.xx:5601
  <Exec>
        if $raw_event =~ s/^(\S+)//
        {
          $HostIP = $Hostname;
          $HostIP = name_to_ipaddr($Hostname);
          if not defined $Hostname $HostIP= $Hostname;

           $raw_event = $HostIP + $raw_event;
        }
    to_json();
   </Exec>

</Output>

<Route eventlog>
  Path      in_eventlog, tcp => out_eventlog
</Route>
```

Alert Types Received

Data from NXlog will undergo an identical process of normalization and enrichment. This ensures that it will provide the same level of alert coverage as our existing Windows sensor.

The following alert types are received when using this NXLog configuration:

- Abnormal Parent / Child Process
- Backup Catalogs Deleted by Ransomware
- Bad Reputation Login
- Command Anomaly
- Encoded PowerShell
- External Account Login Failure Anomaly

- External Brute-Forced Successful User Login
- External Credential Stuffing
- External Password Spraying
- External RDP BlueKeep
- External RDP Suspicious Outbound
- External User Login Failure Anomaly
- Hydra Password Guessing Hack Tool
- Impossible Travel Anomaly
- Internal Account Login Failure Anomaly
- Internal Brute-Forced Successful User Login
- Internal Credential Stuffing
- Internal Password Spraying
- Internal RDP BlueKeep
- Internal RDP Suspicious Outbound
- Internal User Login Failure Anomaly
- Login Time Anomaly
- Malware on Disk
- Mimikatz Credential Dump
- Mimikatz DCSync
- Password Cracking with Hashcat
- Password Spraying Attempts Using Dsacls
- Potentially Malicious Windows Event
- PowerShell Remote Access
- Process Anomaly
- RDP Port Opening
- RDP Registry Modification
- RDP Reverse Tunnel
- RDP Session Hijacking
- RDP Settings Hijacking
- RDP Suspicious Logon
- RDP Suspicious Logon Attempt
- Scanner Reputation Anomaly
- Sensitive Windows Active Directory Attribute Modification
- Sensitive Windows Network Share File or Folder Accessed
- SMB Impacket Lateralization
- SMB Specific Service Installation
- SMB Suspicious Copy
- Suspicious Access Attempt to Windows Object
- Suspicious Activity Related to Security-Enabled Group
- Suspicious Connection to Another Process
- Suspicious Handle Request to Sensitive Object
- Suspicious Powershell Script
- Suspicious Process Creation Commandline
- Suspicious Windows Active Directory Operation
- Suspicious Windows Logon Event
- Suspicious Windows Process Creation

- Suspicious Windows Service Installation
- Uncommon Process Anomaly
- User Asset Access Anomaly
- User Login Location Anomaly
- User Process Usage Anomaly
- Volume Shadow Copy Deletion via VssAdmin
- Volume Shadow Copy Deletion via WMIC

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